

TSCR SERIES

MODEL

TSCR-05E

The TSCR-05E collaborative robot brings automation to workspaces shared with people. Harmonic-drive joints, hand-guided teaching and safety-rated monitored stop let it run fence-free machine tending, assembly, testing and packaging cells. A compact 1052 mm build height and 138 × 138 mm base make it easy to place beside operators on existing benches.

6-Axis	Repeatability	Texsonics RC	Site-ready
Payload 5 kg · Reach 800 mm	±0.03 mm positional accuracy	In-house controller, drives and teach pendant	Floor / Wall / Ceiling mount, factory calibrated

01

SPECIFICATIONS

MECHANICAL		ENVIRONMENT / CONTROL	
AXES	6	HEIGHT	1052 mm
PAYLOAD	5 kg	BASE SIZE	138 × 138 mm
MAX REACH	800 mm	TEACHING	Hand guiding + teach pendant
REPEATABILITY	±0.03 mm	SAFETY	Collaborative operation per ISO/TS 15066 design intent
JOINTS	Harmonic drive, force-limited	PROGRAMMING	Graphical HMI / Offline CAM

02 OVERVIEW

WORKS WITH PEOPLE, NOT BEHIND FENCES.

The TSCR-05E is a collaborative robot with force-limited harmonic-drive joints, hand-guided teaching and safety-rated monitored stop. Deployable on existing benches in days, not months.

01**HAND-GUIDED TEACHING**

Move the arm through the motion and record waypoints — no programming language required.

02**FORCE-LIMITED JOINTS**

Every joint monitors torque and halts on unexpected contact, per the design intent of ISO/TS 15066.

03**SMALL FOOTPRINT**

A 800 mm reach in a 138 × 138 mm base fits beside operators on existing lines.

03 PRODUCT

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WORKING RANGE

MOTION — GROUP A

J1 — BASE ROTATION	±170°
J2 — SHOULDER	±170°
J3 — ELBOW	±160°

MOTION — GROUP B

J4 — WRIST 1	±170°
J5 — WRIST 2	±170°
J6 — TOOL ROTATION	±175°

05

CONTROL & INTEGRATION

CONTROLLER

Texsonics RC series
in-house motion controller,
dual-kernel real-time OS

TEACH PENDANT

11.5" touch tablet
drag-teaching, Wi-Fi,
on-device offline
programming

FIELDBUS

**EtherCAT / Modbus
TCP**
industrial fieldbus,
dual-protocol support

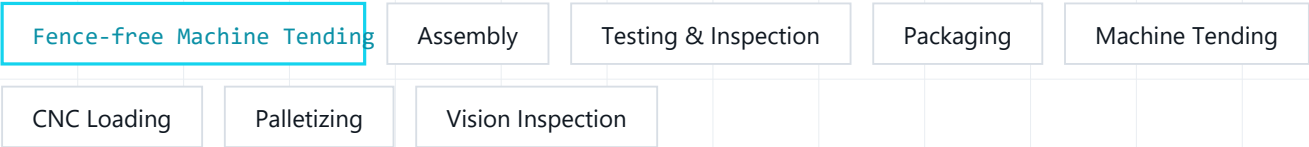
SOFTWARE

TEXCAM offline CAM
toolpath generation, digital
twin simulation, vision

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APPLICATIONS

Deployable across the primary applications the TSCR-05E was engineered for — full production-line integration with the Texsonics control stack.



07

BUILT FOR THE OPERATOR

- | | |
|----------------------------|---------------------------|
| ■ Live 3D position display | ■ I/O monitor |
| ■ Online command execution | ■ Error diagnostics |
| ■ Position editor | ■ Dynamic teach-in |
| ■ Variable monitor | ■ Digital twin simulation |

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TSCR-05E

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TSCR SERIES
6-AXIS · PAYLOAD 5 kg · REACH 800 mm

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